

A comparative study of rule-based, machine learning and large language model approaches in automated writing evaluation (AWE)

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Abstract

Automated Writing Evaluation (AWE) tools have proved beneficial to writing development. Research on AWE methods is essential for improving tool performance and further comparative studies are needed as new methods emerge. This study examines the performance of several AWE approaches, comparing rule-based and statistical methods, machine learning (ML) models, and a large language model (LLM). These three AWE methods were applied to a representative sample of academic essays from the TOEFL11 dataset to compare their assessment performance. Results show that the selected LLM, GPT-4, outperformed the other two approaches in terms of QWK and Pearson's correlation coefficient, while the Support Vector Machine (SVM) model in the ML approach had the highest accuracy and the lowest mean absolute error. This paper provides a detailed comparison of these three approaches and discusses implications for educational practice and future research around AWE.

CCS Concepts

• **Applied computing** → **Computer-assisted instruction**; **E-learning**; • **Computing methodologies** → **Natural language processing**; **Machine learning approaches**.

Keywords

Rule-based method, machine learning, large language model, automated writing evaluation, automated essay scoring, generative AI

ACM Reference Format:

Steven Yeung. 2025. A comparative study of rule-based, machine learning and large language model approaches in automated writing evaluation (AWE). In *LAK25: The 15th International Learning Analytics and Knowledge Conference (LAK 2025)*, March 03–07, 2025, Dublin, Ireland. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706468.3706566>



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LAK 2025, Dublin, Ireland

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ACM ISBN 979-8-4007-0701-8/25/03

<https://doi.org/10.1145/3706468.3706566>

1 Introduction

Writing is a fundamental skill in both academic and professional settings, yet it poses significant challenges to students and professionals at different levels. Despite a massive amount of research on writing and its instruction, the complex nature of writing makes it a difficult skill to teach and assess, requiring substantial time and resources that may not be readily available in many educational settings. The development of Automated Writing Evaluation (AWE) tools is thus considered beneficial. AWE tools provide immediate feedback to writers while also alleviating the assessment burden on instructors. Some studies have recorded the positive impact of AWE tools on the quality of writing [7, 13, 23] as well as citing increases in writer self-efficacy and motivation [20, 21]. The value of these tools lies not only in their efficiency but also in the ability to provide consistent and objective feedback, a challenge often encountered in traditional assessments.

Developments in AWE technology have evolved significantly over the past few decades [9]. To date, there have been three main AWE approaches, namely, rule-based methods, machine learning (ML) models and large language models (LLMs). The earliest developed approaches are predominantly rule and statistics-based, relying on fixed sets of predefined rules, heuristics and indices to identify writing errors and suggest corrections [1]. These are somewhat limited by the rigidity of their rules and may not work well with nuances and variability that are inherent in different types of writing. Later developments with ML have prompted researchers and developers to adopt ML models in AWE system design [12], as these can offer more flexibility and accuracy through algorithms that learn from large datasets of annotated texts. Most recently developed are the LLMs, such as GPT-4. These models can evaluate writing based on vast amounts of data and complex neural network architectures, further enhancing AWE systems. The three approaches listed above each have their strengths and weaknesses in evaluating different writing and linguistic features.

It is therefore of interest to understand and compare the performance of these different approaches. Although some studies have compared the performance of different AWE approaches and addressed limitations for future development [8, 16], the performance of newer models remains under-researched. Such comparative analyses are essential not only for pedagogical purposes but also practical applications, as they help to determine the most suitable AWE methods for assessing different aspects of writing. This study aims to evaluate the effectiveness of traditional rule-based, statistical

and ML approaches against an LLM in AWE, in order to understand the potential for more robust and accurate essay scoring through the integration of multiple methods. This study contributes to the ongoing discussion about the role of artificial intelligence in language assessment and offers insights into future development and implementation of AWE technologies.

2 Related works

2.1 Traditional approaches to AWE (Rule-based and ML approaches)

As the earliest-developed method in AWE, rule-based approaches employ a set of explicit rules and statistical techniques to evaluate essay quality. The idea of rule-based assessment was established as early as the 1960s [14], subsequently evolving into the Project Essay Grade (PEG) system. The underlying assumption of this method is that essay quality can be represented by measurable characteristics. This involves the identification of trins, intrinsic characteristics that human graders seek when evaluating essays. Due to the unobservable nature of trins, quantifiable measures, known as proxes, are approximated using computer programs. For example, essay length measured by total number of words is the prox for the trin of fluency. The emphasis on surface-level features in early AWE systems makes them susceptible to cheating strategies (e.g. the addition of punctuation) that can improve essay ratings without enhancing content [4]. Other natural language processing tools were later developed to address more sophisticated linguistic features that could predict essay scores. Coh-Metrix is one of the comprehensive tools that examined lexical sophistication, syntactic complexity, and text cohesion. Crossley and McNamara [3] found that higher-rated argumentative essays were characterised by higher cohesion and coherency. Tool for the Automatic Analysis of Lexical Sophistication (TAALES 2.0) is another example, designed to assess lexical sophistication using statistics from various corpora. Kyle et al. [10] showed a positive relationship between lexical sophistication and proficiency in writing. Ten of their indices were able to explain 58.0% of the variance in the actual scores. Crossley et al. [2] also found significant negative correlation between grammatical and mechanical errors in GAMET and writing scores by human raters.

Another method in AWE is ML approaches, which employ statistical algorithms to learn scoring criteria without setting pre-defined rules. ML mechanisms can be broadly classified into two main types: unsupervised and supervised. The unsupervised mechanism does not require the use of training data. The goal is to categorise an essay into classes using similarity method and clustering techniques. In contrast, the supervised mechanism utilises evaluation datasets to train the algorithm. Its goal is to predict the score of an essay using regression techniques. Ramesh and Sanampudi [16] categorised the supervised ML techniques of AWE into 4 groups: regression-based models, classification-based models (e.g. [15]), neural networks (e.g. [17]), and the ontology-based approach (e.g. [24]). Their study compared the prediction performance of the first three categories of supervised ML. Poor performance of cohesion and coherence was observed in all models when only statistical features about frequency of word occurrences were used in training. If unidirectional semantic information was included in training, neural network would yield better performance than regression and classification models.

2.2 The potential of an LLM approach

LLMs are artificial intelligence models trained on vast datasets and sophisticated algorithms. Different from the previous two methods, LLMs have the potential to contribute to AWE given their ability to receive prompts and respond. Some studies have begun to examine the performance of LLMs in AWE. Mizumoto and Eguchi [12] investigated the use of the LLM GPT-3 in evaluating the ETS Corpus of Non-Native Written English (TOEFL11) and compared these scores to benchmark levels. Their study showed the model's ability to align generally well with human scoring benchmarks, though with some score overlaps and limitations in exact score matching, as indicated by a quadratic weighted kappa (QWK) of 0.682. Similarly, Li et al. [11] examined the effectiveness of an upgraded version of GPT, GPT-4, in assessing the quality of young students' essay revisions. It was found that while GPT-4 showed potential for evaluating revision quality, particularly with the use of detailed rubrics outlining common revision patterns, the use of Chain-of-Thought (CoT) prompting did not significantly enhance performance. The initial zero-shot prompting experiment yielded an agreement rate of 52.00% and a QWK of 0.60, indicating moderate alignment with human raters. Subsequent experiments that included examples of successful revisions improved the exact agreement rate to 54.50%, but the QWK remained unchanged. Notably, GPT-4 showed higher agreement with human scoring for Level 1 drafts, which focused on adding evidence, compared to more complex Level 2 and Level 3 revisions that required elaboration and explanation. This suggests that GPT-4 is better suited for straightforward evaluation tasks, while its performance may decline when assessing multifaceted revisions.

2.3 The present study

It can be hypothesised that different AWE approaches perform differently when assessing writing, as indicated by the available research [8, 16]. Nevertheless, there is a lack of research comparing the performance of traditional approaches with the latest LLM methods. Given that LLM models are a relatively recent development, more research is warranted. This study therefore seeks to answer one overarching question:

How does the performance of three AWE approaches, namely (1) rule-based, (2) ML and (3) LLM, differ in predicting essay quality?

3 Methodology

3.1 Dataset

The TOEFL11 dataset was used in this study. It is a rich and well-established corpus of non-native written English, a type of writing commonly used in AWE and other natural language processing studies. The dataset is derived from the Test of English as a Foreign Language (TOEFL), a standardised test for assessing academic English proficiency of non-native speakers. The dataset contains 12,100 essays written by test takers from 11 different native language backgrounds from 2006 to 2007. The essays come from the TOEFL Independent Writing Task, where test takers respond to one of eight prompts, resulting in a variety of topics and writing styles as well as performance levels. Each essay was first graded by trained human raters on a 5-point scale. The sum of the scores was then collapsed into a 3-level proficiency scale categorised into low, mid and high groups, representing low, intermediate and high levels of

language proficiency. This stratification enables us to evaluate how well the selected AWE methods perform across different skill levels. As a result, stratified sampling was applied to the TOEFL11 dataset to ensure a balanced representation of essay quality. In the 10% representative sample, there were 133 low-level essays (11%), 657 medium-level essays (54%), and 420 high-level essays (35%). This distribution reflects the typical spread of essay quality in real-world scenarios, allowing the selected models to learn from a diverse range of writing styles and proficiency levels.

3.2 Feature extraction

We employed a comprehensive set of natural language processing analysis tools to capture the multifaceted nature of essay quality. We opted for tools with relevant features to measure six common areas of writing, namely text cohesion, paragraph organisation, grammar usage, spelling and punctuation accuracy, language and vocabulary choices, and syntactic complexity. The tool Coh-Metrix was chosen for text cohesion, offering insight into how well ideas were connected within the essays. GAMET (Grammar and Mechanics Error Tool) was used to identify grammatical and mechanical errors. TAALES (Tool for the Automated Analysis of Lexical Sophistication) was selected to assess the sophistication of vocabulary choice in the essays. TAASSC (Tool for the Automatic Analysis of Syntactic Sophistication and Complexity) was used to examine the complexity of sentence structures. These four tools collectively extracted 322 linguistic features to measure the six aspects of writing quality listed above. This provided a rich set of data points, allowing us to capture both surface-level features and higher-order linguistic and textual characteristics.

3.3 A rule-based approach

We implemented a rule-based approach to establish a baseline and explore the predictive power of individual features. First, we stratified the sample and used an 80/20 split for the train-test ratio. This resulted in 968 essays in the training set and 242 essays in the test set, including 27 low-level, 131 mid-level and 84 high-level essays. Next, we conducted a correlation analysis to identify the features most strongly associated with essay quality. The correlation of each of the 322 linguistic features with the actual essay levels was calculated. We sorted the features in ascending order if the correlation sign was positive, and descending order if the sign was negative. This allowed us to establish appropriate thresholds for each feature for comparison. In threshold determination, we chose the threshold for each feature based on the distribution of low, mid, and high levels in the training data. The general rule was $low < x \leq medium < y \leq high$, with x and y being the threshold values specific to each feature. As an example, consider the feature 'number of words' by focusing on a small sample of 7 essays with a hypothetical proportion of low, mid, and high quality in the ratio of 1:2:4 (while the ratio of the test set was 27:131:84 in the actual case). In this miniature sample, with word counts sorted in ascending order (100 [low], 200 [mid], 250 [mid], 300 [high], 350 [high], 400 [high], 450 [high]), thresholds x and y were set at 150 and 275 respectively based on the given levels. These thresholds established the following prediction rules: essays with < 150 words were classified as low-level, essays with 150–274 words as mid-level, and essays with ≥ 275 words as high-level. When applying these thresholds to a test case with three essays, all having 100 words,

our model predicted all the essays as low-level, while the actual levels could be low, low, and mid, resulting in an accuracy of 67%. We applied these thresholds to the test set to evaluate the predictive power of each feature independently.

3.4 Supervised ML models

For the ML approach, we used Scikit-learn Python module to evaluate two popular ML algorithms, Random Forest (RF) and Support Vector Machine (SVM). The same 80/20 train-test split was applied for both models to ensure a fair comparison.

In the RF approach, we first used the extremely randomized trees (ExtraTreesClassifier function) for feature selection. The selection of the most important features helped reduce dimensionality and possibly improve model performance. Second, we addressed class imbalance by applying the Synthetic Minority Over-sampling Technique (SMOTE). This data augmentation method involved the creation of synthetic examples of the minority class, which balanced the class distribution and enhanced model performance in later steps. Third, we performed feature scaling by standardising the features to zero mean and unit variance using the StandardScaler function. Fourth, we applied the RandomForestClassifier function for the actual prediction. To tune the hyperparameters, we performed 5-fold cross-validation and selected the best parameters. The model was re-trained on the entire training set and finally evaluated on the test set. In the SVM approach, the data transformation procedures were similar to those of the RF, with one exception. The difference was that the feature selection used the LinearSVC function with L1 penalty to produce sparse solutions and eliminate less important features.

3.5 An LLM approach

For the LLM approach, we chose to evaluate the prediction performance of ChatGPT, a prominent LLM developed by OpenAI [5]. In terms of scoring, we adapted the original 5-point TOEFL rubric into a 3-point scale (low, medium, high) to align with the scoring provided by the given corpus. Low proficiency corresponded to scores ranging from 1.0 to 2.0, medium to scores between 2.5 and 3.5, and high to scores between 4.0 and 5.0. Regarding the ChatGPT configuration, GPT-4 Turbo preview model (gpt-4-1106-preview) was used through the OpenAI API. To mimic the two human raters in the original TOEFL11 dataset, ChatGPT rated each essay twice with different temperature values. It rated once with temperature 0 for more predictable and consistent responses, and again with temperature 0.7 for more variability. We then took the average of these two ratings to produce the final level for each essay. We used a specific system prompt that instructed ChatGPT to act as a rater for a high-stakes English language exam, providing the TOEFL iBT Test Independent Writing Rubrics as a reference (See appendices A and B for the full system prompts and user message).

3.6 Performance evaluation

We used QWK as our primary metric in performance evaluation. QWK is particularly suitable for ordinal classification tasks like essay grading, as it accounts for the degree of disagreement between predicted and actual grades. Additionally, we listed other metrics, namely accuracy, Pearson's correlation coefficient, and mean absolute error. Accuracy shows the proportion of correct matches of

Table 1: Top 5 performing features ranked by QWK

Ranking	Name of feature	Description	Tool	QWK	Accuracy	Pearson's correlation coefficient	Mean absolute error
1	DESWC	Number of words	Coh-Metrix	0.1650	0.4008	-0.5285	0.8636
2	KF_Ncats_CW	Kucera-Francis number of categories for content words	TAALES	0.1469	0.4545	0.2231	0.6529
3	Brysbaert_Concreteness_Combined_AW	Brysbaert concreteness combined – all words (Average combined concreteness score)	TAALES	0.1414	0.4504	0.1325	0.6529
4	misspelling_per_100_words	Misspelling per 100 words	GAMET	0.1399	0.4669	0.4408	0.5950
5	acad_lemma_construction_ttr	Academic lemma construction type-token ratio	TAASSC	0.1333	0.4421	0.0621	0.6860

Table 2: Performance metrics of RF and SVM

	QWK	Accuracy	Pearson's correlation coefficient	Mean absolute error
RF	0.3967	0.6860	0.5779	0.3182
SVM	0.5098	0.7190	0.6337	0.2810

predicted and actual grades, with a higher value indicating better prediction performance. Pearson's correlation coefficient provides the correlation coefficient between predicted and actual grades. Its values are between -1 and +1, with the sign indicating direction of relationship and the magnitude indicating strength of correlation. Mean absolute error is the actual difference between the predicted and actual levels that only shows the strength of relationship.

4 Results

4.1 The rule-based approach

The rule-based approach served as a baseline and helped identify potentially important features. Table 1 shows that the best-performed feature was the number of words ('DESWC') in an essay with QWK=0.1650, suggesting that essay length had a slight correlation with essay quality. The second and third best-performed features came from TAALES, implying the importance of lexical sophistication in essay assessment. The diversity of content words ('KF_Ncats_CW') and the use of concrete language ('Brysbaert_Concreteness_Combined_AW') were predictive of quality. The fourth feature, misspelling per 100 words ('misspelling_per_100_words'), showed that fewer spelling errors correlated with higher essay levels. The fifth feature, academic lemma construction type-token ratio ('acad_lemma_construction_ttr'), suggested the positive predictive power of variety of academic vocabulary on essay quality. While these individual features showed some correlation with essay levels, the QWK values are considered relatively low (below 0.2).

4.2 The ML approach

The limited performance of the rule-based approach calls for more complex models in predicting essay quality. In the ML approach, we

used RF and SVM to model essay quality. From Table 2, we observed that SVM outperformed RF in all four metrics. It showed higher accuracy, better agreement with human raters, stronger correlation with actual grades, and lower mean absolute error.

Tables 3-4 list the top 5 features in the trained ML models by importance, measured by mean decrease in impurity in RF and feature permutation in SVM. Number of words ('DESWC') was the only feature that appeared in both the rule-based and two ML models, highlighting the importance of essay length in assessment. In the RF method, the top 4 important features mainly came from Coh-Metrix and GAMET, indicating the significance of cohesion and mechanical accuracy. In the SVM method, the top 5 important features were related to word count and bigram measures. This showed the change in focus to lexical sophistication in this method.

To show the classification performance in each range (high, mid and low), we examined both models using confusion matrices (see Figure 1). Both matrices indicate that both models performed best in the mid-range classifications, while SVM showed better performance in the high range compared to RF. The confusion matrices suggest that both models struggled somewhat with distinguishing between adjacent classes, which is a common challenge in ordinal classification tasks like essay grading.

4.3 The LLM approach

The prediction results of the chosen LLM, ChatGPT, are shown in Table 5. Compared to ML models, ChatGPT classification results were better except for the accuracy metric and mean absolute error. The QWK of ChatGPT was 0.5707, indicating good agreement between predicted and actual grades. The accuracy of 0.5992 meant that 59.92% of predictions were correct. Pearson's correlation coefficient was 0.6373, implying strong positive correlation between

Table 3: Top 5 most important features in RF

Ranking	Name of feature	Description	Tool	Impurity-based feature importance
1	DESWC	Number of words	Coh-Metrix	0.1057
2	misspelling_per_100_words	Misspelling per 100 words	GAMET	0.0520
3	DESSC	Number of sentences	Coh-Metrix	0.0296
4	grammar_per_100_words	Grammatical errors per 100 words	GAMET	0.0241
5	Brown_Freq_AW_log	Brown mean frequency score - all words (log transformed)	TAALES	0.0215

Table 4: Top 5 most important features in SVM

Ranking	Name of feature	Description	Tool	Permutation importance
1	COCA_academic_bi_DP	COCA academic bigram association strength (mean delta P association score)	TAALES	0.0035
2	COCA_Academic_Bigram_Range	COCA academic mean bigram range score	TAALES	0.0031
3	DESWC	Number of words	Coh-Metrix	0.0027
4	BNC_Written_Bigram_Freq_Normed_Log	BNC written bigram mean frequency score (log transformed)	TAALES	0.0027
5	all_av_lemma_freq_type	Average lemma frequency (types only) – all	TAASSC	0.0025

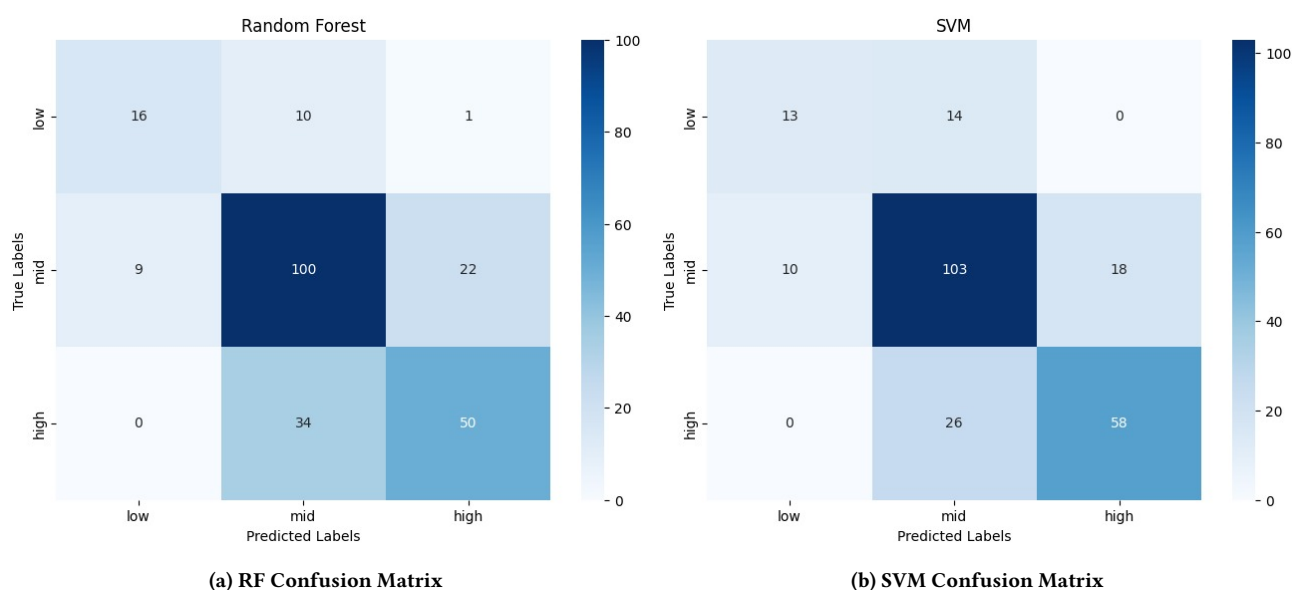


Figure 1: Confusion matrices of RF and SVM models

predicted and actual grades. The mean absolute error was 0.4050, meaning that predictions were off by about 0.41 grade points on average.

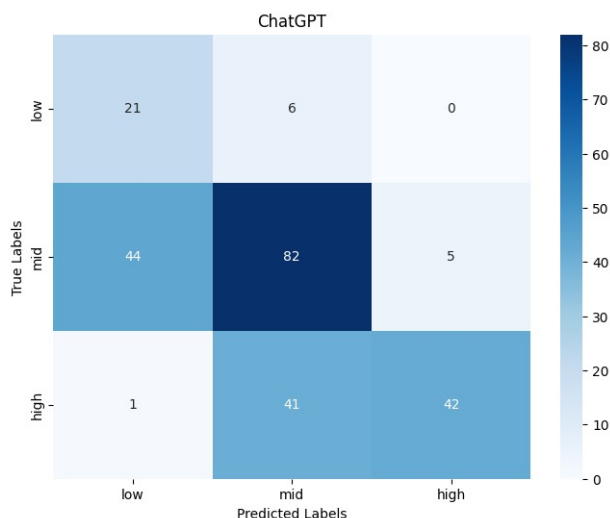
The confusion matrix in Figure 2 shows more misclassifications of high into mid and mid into low. This indicates that ChatGPT performed better in identifying high and low levels, with some confusion in the mid range.

5 Discussion

The findings of this study offer valuable insights into the effectiveness of rule-based, ML and LLM approaches in predicting essay quality. The rule-based and statistical approach served as a baseline. While features such as essay length, lexical sophistication, and spelling accuracy correlated with quality, their QWK values remained low (below 0.2). This suggests that no single feature was sufficient and that a more sophisticated approach is needed to capture the complexities of essay evaluation. For such an approach, one

Table 5: Performance metrics of ChatGPT

	QWK	Accuracy	Pearson’s correlation coefficient	Mean absolute error
ChatGPT	0.5707	0.5992	0.6373	0.4050

**Figure 2: Confusion matrix of ChatGPT**

could seek to combine multiple features or use more advanced algorithms able to capture the relationships between various linguistic features and overall essay quality. However, developing more sophisticated and comprehensive rule-based systems and supervised ML models requires expert knowledge to engineer features and is therefore more resource-intensive [6]. As for the ML approach, the SVM model had the highest accuracy (0.7190) and the lowest mean absolute error (0.2810), whereas the LLM outperformed the other two approaches in terms of QWK (0.5707) and Pearson’s correlation coefficient (0.6373). These results demonstrate the potential of ML-based and LLM-based AWE methods while also highlighting areas for potential improvement, such as the need to better differentiate between adjacent grade levels. In the case of LLMs, further finetuning and prompting techniques may enhance performance [19].

This study also has implications for educational practice. First, features such as essay and sentence length, word choice and spelling that were found to be better predictors of scores (as measured using the TOEFL dataset) could be prioritised in the writing classroom to improve student performance. Second, due to their accessibility and strengths in certain areas of AWE, LLMs could be employed as a complementary tool in the writing process to offer support in formative and low-stakes assessment. Although the scores given by LLMs are difficult to explain due to their ‘black-box’ nature, students could benefit from receiving qualitative feedback. Many recent studies have already suggested the benefits of using LLMs in the writing classroom (e.g. [18, 22]).

6 Limitations and future work

This study has several limitations. First, it focused exclusively on TOEFL academic essays, which may not represent a full diversity of writing styles and genres. Certain writing features could be more significant in this specific context, potentially skewing results. Another limitation is the use of basic linguistic features in ML models, excluding advanced text representations such as TF-IDF or complex embeddings. This restriction likely hindered the models’ ability to capture nuanced aspects of essay quality. In contrast, GPT’s pre-training on extensive text data enables it to utilise rich, contextual language representations, resulting in more effective evaluations. Lastly, the selected ML models were trained on only 10% of the available essays, using an 80/20 train-test split. This limited training data may have put the ML models at a disadvantage compared to GPT, which benefits from extensive pre-training on diverse texts.

For future work, integrating advanced deep learning architectures and exploring sophisticated text representation methods could enhance ML models’ ability to capture relevant linguistic features for essay evaluation. In addition, more detailed comparative studies could offer deeper insights into specific conditions or essay types where one method may outperform others. Future studies could compare more LLMs and their scoring capabilities in different genres and different settings (e.g. with different prompts). Such studies could pinpoint precise strengths and weaknesses and contribute to more targeted improvements of AWE systems.

7 Conclusion

This study on AWE methods has yielded telling results, showing clear differences in performance among the three approaches investigated: rule-based, ML, and LLM approaches. These results offer insight into the most effective methods for reliable and accurate essay scoring. While this study demonstrates the superiority of GPT-based scoring in QWK and Pearson’s correlation coefficient, it also points to the potential of advanced ML approaches in AWE. As LLMs continue to develop, their potential is also expected to grow. The implementations of AWE systems in the future will likely involve an informed combination of different computational methods to balance performance with practical considerations. As AWE methods continue to evolve, maintaining a focus on accuracy and reliability, interpretability, and pedagogical value will be crucial in developing systems that can enhance learning experiences.

Acknowledgments

This study was inspired by my professional exchange with Professor Mutlu Cukurova, who shared one of his co-authored papers on predicting challenge moments from students’ discourse. I would like to express my appreciation to Professor Cukurova for his insights and to Tommy Tam and Maggie Fok for their technical support. This

study was part of a larger project on developing an automated writing evaluation platform funded by the Teaching Development and Language Enhancement Grant (TDLEG) (2022–25) at The Chinese University of Hong Kong.

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Appendices

A ChatGPT system message

You are a rater for writing responses on a high-stakes English language exam for second language learners. You will be provided with a prompt and the test-taker's response. Ratings are based on the TOEFL iBT Test Independent Writing Rubrics. Each rating should be one of the following: 5, 4, 3, 2, 1, 0.

An essay at level 5 largely accomplishes all of the following:

- effectively addresses the topic and task
- is well organized and well developed, using clearly appropriate explanations, exemplifications, and/or details
- displays unity, progression, and coherence
- displays consistent facility in the use of language, demonstrating syntactic variety, appropriate word choice, and idiomatcity, though it may have minor lexical or grammatical errors

An essay at level 4 largely accomplishes all of the following:

- addresses the topic and task well, though some points may not be fully elaborated
- is generally well organized and well developed, using appropriate and sufficient explanations, exemplifications, and/or details
- displays unity, progression, and coherence, though it may contain occasional redundancy, digression, or unclear connections
- displays facility in the use of language, demonstrating syntactic variety and range of vocabulary, though it will probably have occasional noticeable minor errors in structure, word form, or use of idiomatic language that do not interfere with meaning

An essay at level 3 is marked by one or more of the following:

- addresses the topic and task using somewhat developed explanations, exemplifications, and/or details
- displays unity, progression, and coherence, though connection of ideas may be occasionally obscured
- may demonstrate inconsistent facility in sentence formation and word choice that may result in lack of clarity and occasionally obscure meaning
- may display accurate but limited range of syntactic structures and vocabulary

An essay at level 2 may reveal one or more of the following weaknesses:

- limited development in response to the topic and task

- inadequate organization or connection of ideas
- inappropriate or insufficient exemplifications, explanations, or details to support or illustrate generalizations in response to the task
- a noticeably inappropriate choice of words or word forms
- an accumulation of errors in sentence structure and/or usage

An essay at level 1 is seriously flawed by one or more of the following weaknesses:

- serious disorganization or underdevelopment
- little or no detail, or irrelevant specifics, or questionable responsiveness to the task
- serious and frequent errors in sentence structure or usage

An essay at level 0 merely copies words from the topic, rejects the topic, or is otherwise not connected to the topic, is written in a foreign language, consists of keystroke characters, or is blank.

You should reply to each response with just your rating: do not explain or justify it.

B ChatGPT user message

Prompt: ""
{essay_prompt}
""

Response: ""
{essay_response}
""